

AWS Multi-Architecture C Application Deployment

Work Documentation with Screenshots

Amazon ECR • Amazon S3 • EC2 • IAM • Docker • GCC

Prepared from the provided AWS Console and EC2 terminal screenshots

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1. Create Amazon ECR Private Repository

Created the private Amazon Elastic Container Registry repository c-multiarch-app for storing the container image.

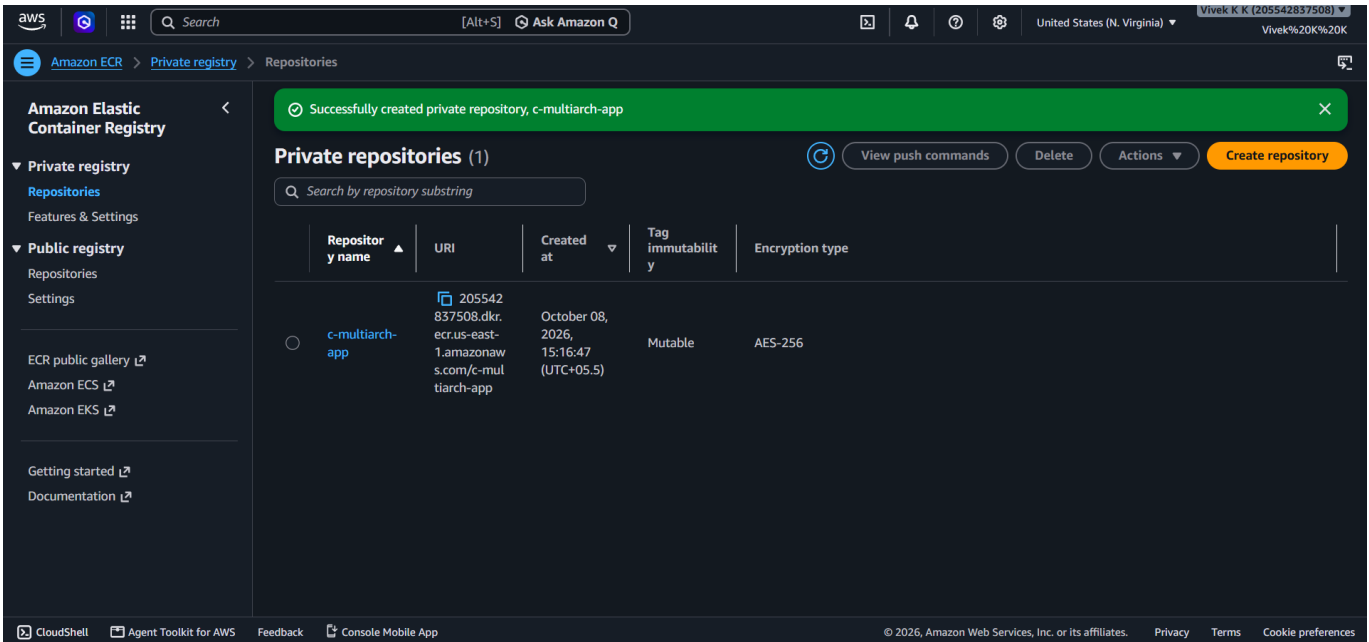


Figure 1. 1. Create Amazon ECR Private Repository

2. Create Amazon S3 Artifact Bucket

Created the S3 bucket my-c-app-artifacts-2026 in us-east-1 for application artifacts and binaries.

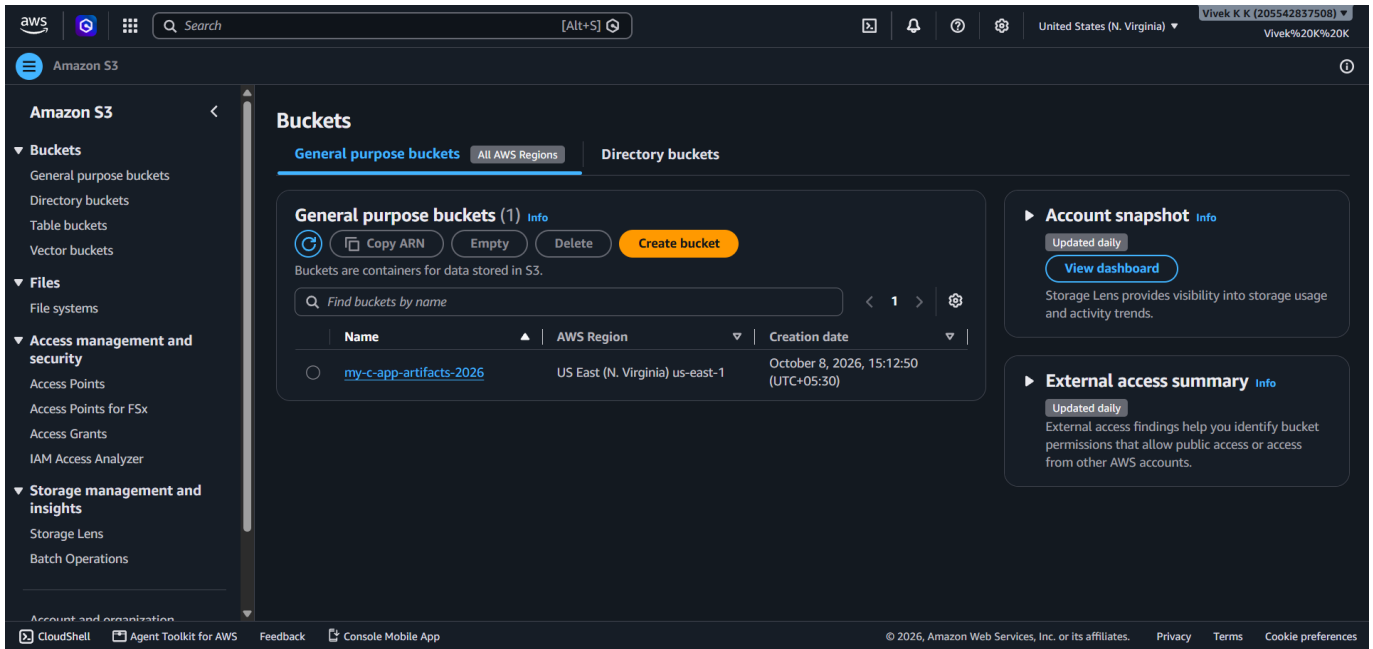


Figure 2. 2. Create Amazon S3 Artifact Bucket

3. Launch Multi-Architecture EC2 Instances

Verified two running EC2 instances: an x86_64 c-app-x86 instance and an ARM64/Graviton c-app-graviton instance.

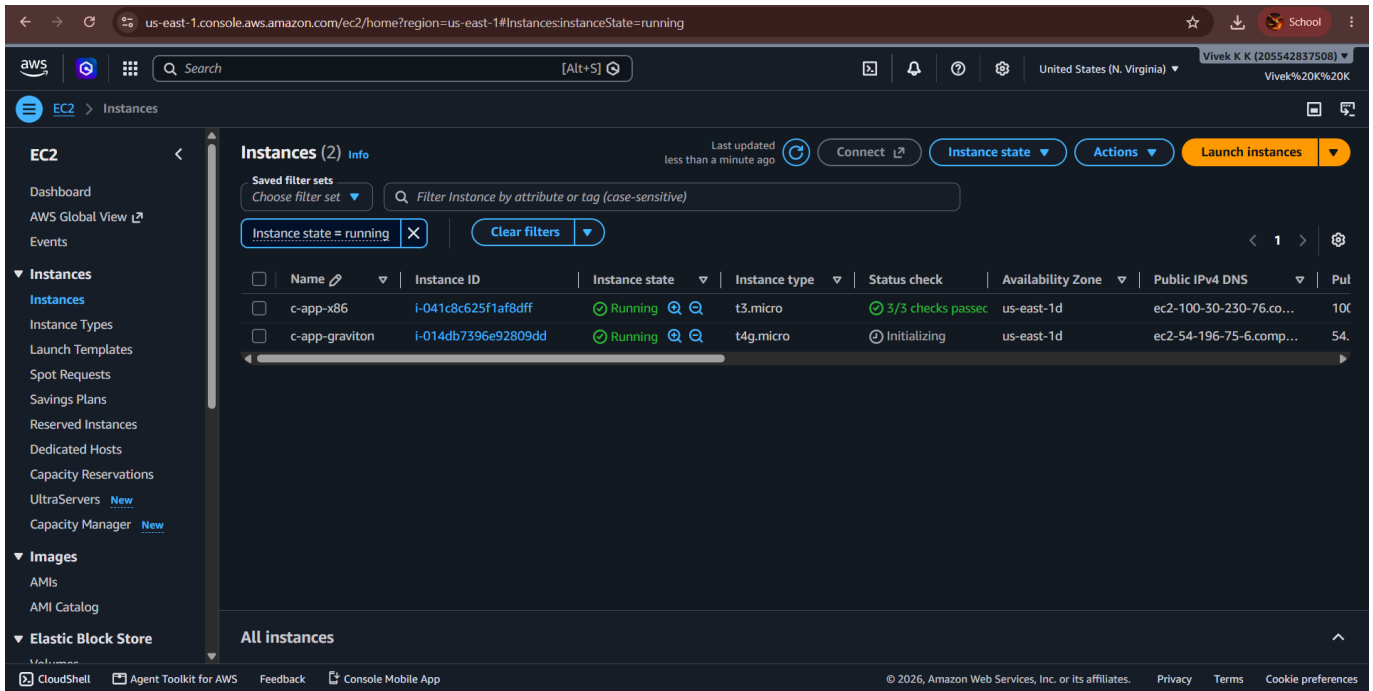


Figure 3. 3. Launch Multi-Architecture EC2 Instances

4. Configure EC2 Security Group for SSH

Configured the EC2 security group with an SSH rule on TCP port 22 restricted to the specified source IP.

The screenshot displays the AWS Management Console interface for configuring an EC2 Security Group. The left sidebar shows the navigation menu with categories like Images, Elastic Block Store, and Network & Security. The main content area is titled 'Security Groups (1/2)' and shows a list of security groups. The selected security group, 'sg-0ca95254e21f6c2ab' (launch-wizard-1), is highlighted. Below the list, the 'Inbound rules' tab is active, showing a single rule for SSH access on TCP port 22, restricted to the source IP 121.200.55.210/32.

Name	Security group ID	Security group name	VPC ID	Description
-	sg-0ca95254e21f6c2ab	launch-wizard-1	vpc-06c368c2c0eefa623	launch-wizard-1 cre...
-	sg-07343acd696350a51	default	vpc-06c368c2c0eefa623	default VPC security

Security group rule ID	IP version	Type	Protocol	Port range	Source	D
sgr-062789f643ae4c129	IPv4	SSH	TCP	22	121.200.55.210/32	-

Figure 4. 4. Configure EC2 Security Group for SSH

5. Verify x86_64 EC2 Architecture

Connected to the c-app-x86 instance and verified the machine architecture using `uname -m`, returning `x86_64`.

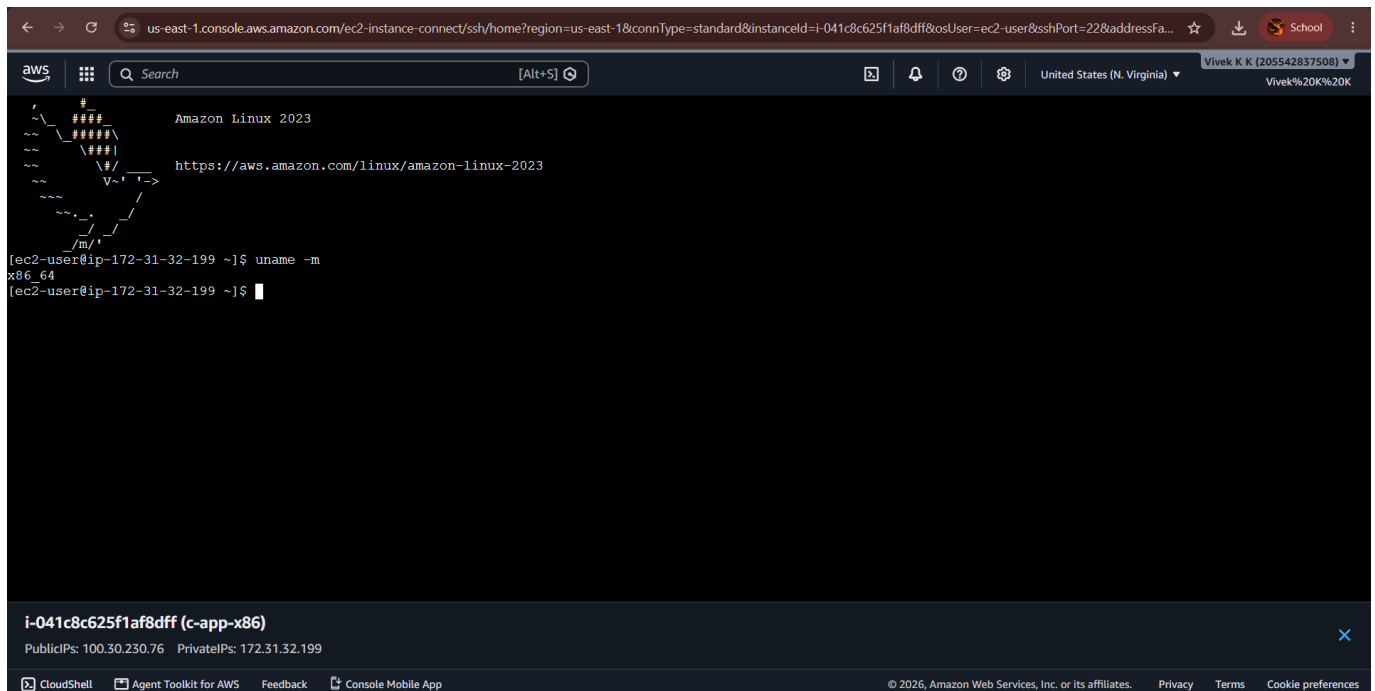
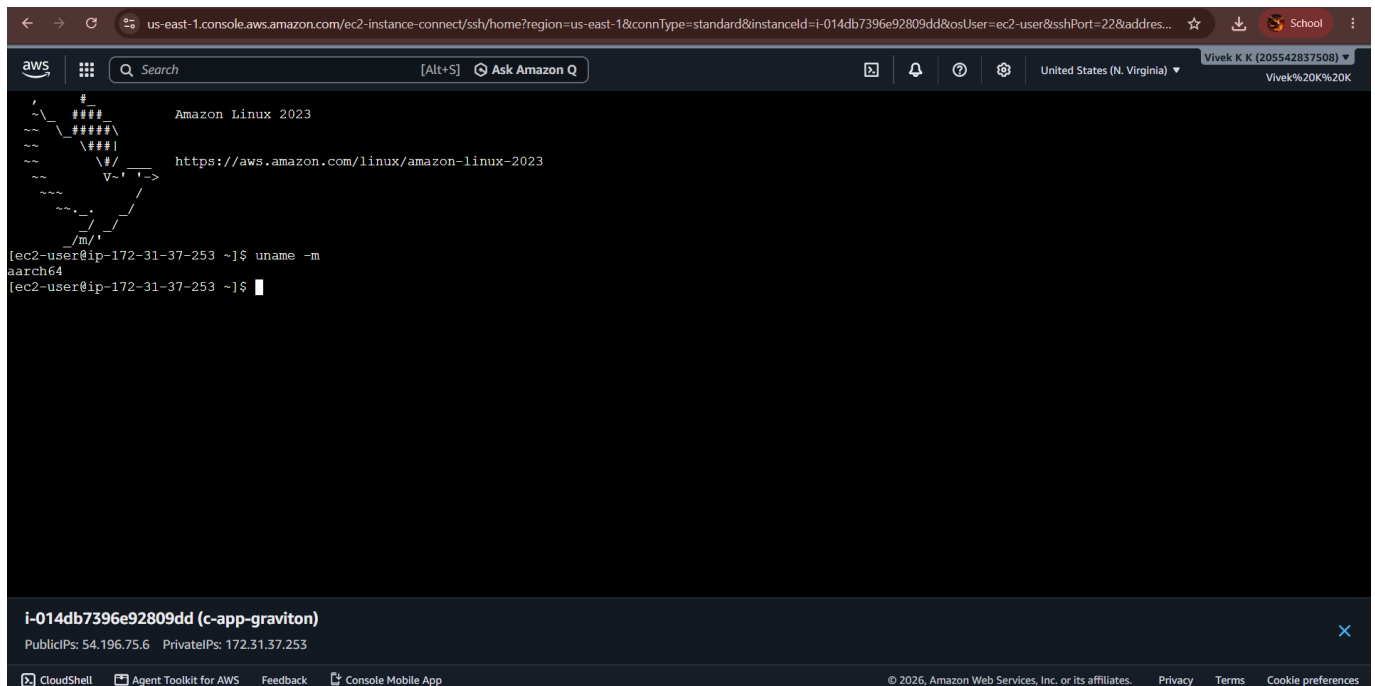


Figure 5. 5. Verify x86_64 EC2 Architecture

6. Verify ARM64 EC2 Architecture

Connected to the c-app-graviton instance and verified the machine architecture using `uname -m`, returning `aarch64`.



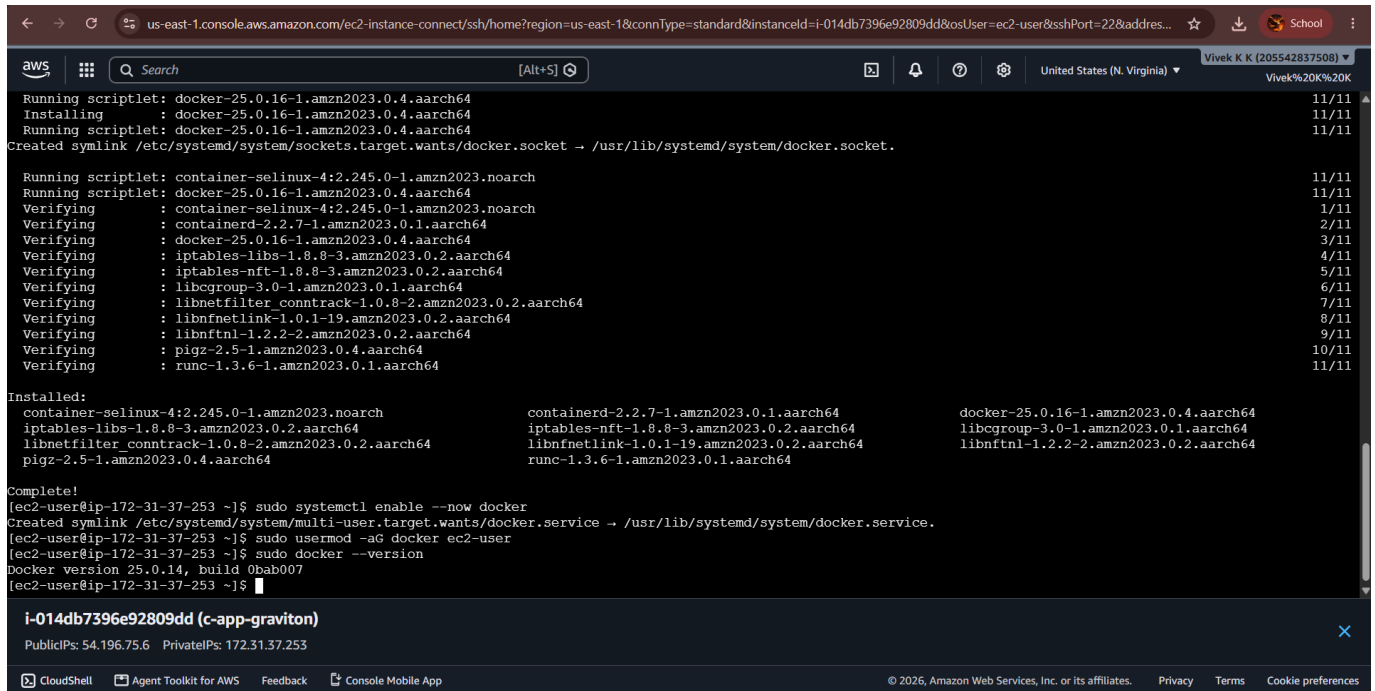
The screenshot shows an AWS CloudShell terminal window. The top bar displays the AWS logo, a search bar, and the region 'United States (N. Virginia)'. The terminal output shows the Amazon Linux 2023 logo and the URL <https://aws.amazon.com/linux/amazon-linux-2023>. The user runs the command `uname -m`, which returns `aarch64`. The bottom of the terminal shows the instance ID `i-014db7396e92809dd` and the name `(c-app-graviton)`. The footer of the terminal window includes links to CloudShell, Agent Toolkit for AWS, Feedback, and Console Mobile App, along with copyright information for Amazon Web Services, Inc.

```
us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=us-east-1&connType=standard&instanceId=i-014db7396e92809dd&osUser=ec2-user&sshPort=22&addres...  
aws [Alt+S] Ask Amazon Q United States (N. Virginia) Vivek K K (205542837508) Vivek%20K%20K  
Amazon Linux 2023  
https://aws.amazon.com/linux/amazon-linux-2023  
[ec2-user@ip-172-31-37-253 ~]$ uname -m  
aarch64  
[ec2-user@ip-172-31-37-253 ~]$  
i-014db7396e92809dd (c-app-graviton)  
PublicIPs: 54.196.75.6 PrivateIPs: 172.31.37.253  
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```

Figure 6. 6. Verify ARM64 EC2 Architecture

7. Install and Enable Docker on ARM64

Installed Docker and its dependencies on the ARM64 instance and enabled the Docker service with systemd.



```
us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=us-east-1&connType=standard&iinstanceId=i-014db7396e92809dd&osUser=ec2-user&sshPort=22&addres...
Vivek K K (205542837508)
Vivek%20K%20K

Running scriptlet: docker-25.0.16-1.amzn2023.0.4.aarch64 11/11
Installing      : docker-25.0.16-1.amzn2023.0.4.aarch64 11/11
Running scriptlet: docker-25.0.16-1.amzn2023.0.4.aarch64 11/11
Created symlink /etc/systemd/system/sockets.target.wants/docker.socket → /usr/lib/systemd/system/docker.socket.

Running scriptlet: container-selinux-4:2.245.0-1.amzn2023.noarch 11/11
Running scriptlet: docker-25.0.16-1.amzn2023.0.4.aarch64 11/11
Verifying      : container-selinux-4:2.245.0-1.amzn2023.noarch 1/11
Verifying      : containerd-2.2.7-1.amzn2023.0.1.aarch64 2/11
Verifying      : docker-25.0.16-1.amzn2023.0.4.aarch64 3/11
Verifying      : iptables-libs-1.8.8-3.amzn2023.0.2.aarch64 4/11
Verifying      : iptables-nft-1.8.8-3.amzn2023.0.2.aarch64 5/11
Verifying      : libcgrou-3.0-1.amzn2023.0.1.aarch64 6/11
Verifying      : libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.aarch64 7/11
Verifying      : libnftnl-1.0.1-19.amzn2023.0.2.aarch64 8/11
Verifying      : libnftnl-1.2.2-2.amzn2023.0.2.aarch64 9/11
Verifying      : pigz-2.5-1.amzn2023.0.4.aarch64 10/11
Verifying      : runc-1.3.6-1.amzn2023.0.1.aarch64 11/11

Installed:
container-selinux-4:2.245.0-1.amzn2023.noarch      containerd-2.2.7-1.amzn2023.0.1.aarch64      docker-25.0.16-1.amzn2023.0.4.aarch64
iptables-libs-1.8.8-3.amzn2023.0.2.aarch64      iptables-nft-1.8.8-3.amzn2023.0.2.aarch64      libcgrou-3.0-1.amzn2023.0.1.aarch64
libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.aarch64      libnftnl-1.0.1-19.amzn2023.0.2.aarch64      libnftnl-1.2.2-2.amzn2023.0.2.aarch64
pigz-2.5-1.amzn2023.0.4.aarch64                  runc-1.3.6-1.amzn2023.0.1.aarch64

Complete!
[ec2-user@ip-172-31-37-253 ~]$ sudo systemctl enable --now docker
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /usr/lib/systemd/system/docker.service.
[ec2-user@ip-172-31-37-253 ~]$ sudo usermod -aG docker ec2-user
[ec2-user@ip-172-31-37-253 ~]$ sudo docker --version
Docker version 25.0.14, build 0bab007
[ec2-user@ip-172-31-37-253 ~]$
```

i-014db7396e92809dd (c-app-graviton)

PublicIPs: 54.196.75.6 PrivateIPs: 172.31.37.253

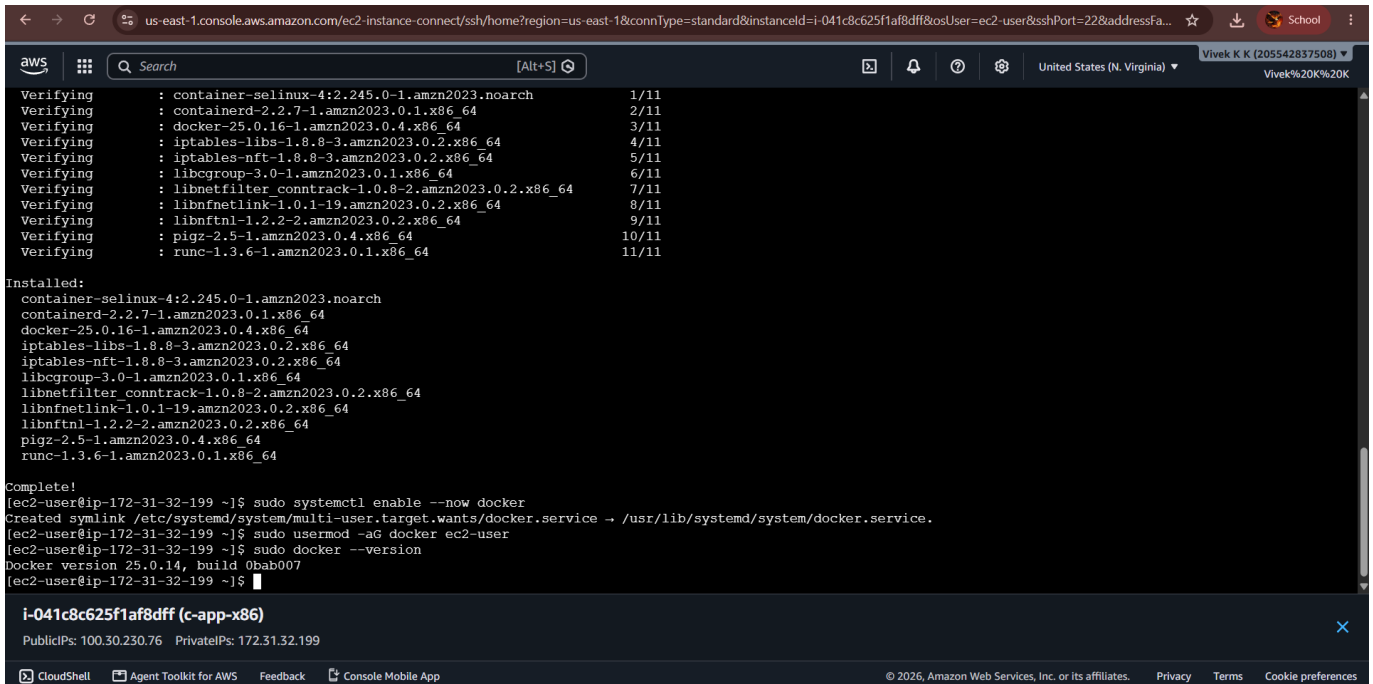
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Figure 7. 7. Install and Enable Docker on ARM64

8. Install and Enable Docker on x86_64

Installed Docker and its dependencies on the x86_64 instance and enabled the Docker service with systemd.



```
us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=us-east-1&connType=standard&instanceId=i-041c8c625f1af8dff&osUser=ec2-user&sshPort=22&addressFa...
aws Search [Alt+S]
United States (N. Virginia)
Vivek K K (205542837508)
Vivek%20K%20K

Verifying : container-selinux-4:2.245.0-1.amzn2023.noarch 1/11
Verifying : containerd-2.2.7-1.amzn2023.0.1.x86_64 2/11
Verifying : docker-25.0.16-1.amzn2023.0.4.x86_64 3/11
Verifying : iptables-libs-1.8.8-3.amzn2023.0.2.x86_64 4/11
Verifying : iptables-nft-1.8.8-3.amzn2023.0.2.x86_64 5/11
Verifying : libcgrou-3.0-1.amzn2023.0.1.x86_64 6/11
Verifying : libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64 7/11
Verifying : libnftnl-1.0.1-19.amzn2023.0.2.x86_64 8/11
Verifying : libnftnl-1.2.2-2.amzn2023.0.2.x86_64 9/11
Verifying : pigz-2.5-1.amzn2023.0.4.x86_64 10/11
Verifying : runc-1.3.6-1.amzn2023.0.1.x86_64 11/11

Installed:
container-selinux-4:2.245.0-1.amzn2023.noarch
containerd-2.2.7-1.amzn2023.0.1.x86_64
docker-25.0.16-1.amzn2023.0.4.x86_64
iptables-libs-1.8.8-3.amzn2023.0.2.x86_64
iptables-nft-1.8.8-3.amzn2023.0.2.x86_64
libcgrou-3.0-1.amzn2023.0.1.x86_64
libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64
libnftnl-1.0.1-19.amzn2023.0.2.x86_64
libnftnl-1.2.2-2.amzn2023.0.2.x86_64
pigz-2.5-1.amzn2023.0.4.x86_64
runc-1.3.6-1.amzn2023.0.1.x86_64

Complete!
[ec2-user@ip-172-31-32-199 ~]$ sudo systemctl enable --now docker
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service -> /usr/lib/systemd/system/docker.service.
[ec2-user@ip-172-31-32-199 ~]$ sudo usermod -aG docker ec2-user
[ec2-user@ip-172-31-32-199 ~]$ sudo docker --version
Docker version 25.0.14, build 0bab007
[ec2-user@ip-172-31-32-199 ~]$
```

i-041c8c625f1af8dff (c-app-x86)
PublicIPs: 100.30.230.76 PrivateIPs: 172.31.32.199

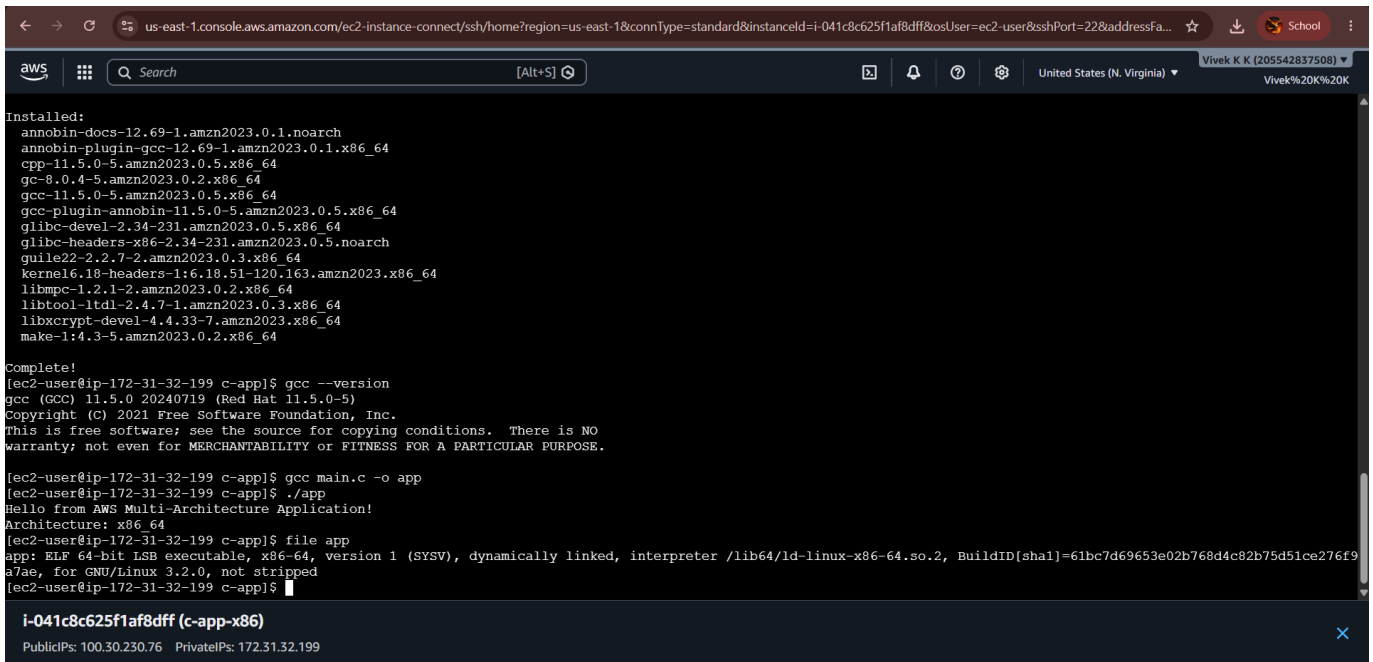
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Figure 8. 8. Install and Enable Docker on x86_64

9. Install GCC and Build the C Application on x86_64

Installed GCC development tools, compiled main.c into the app executable, ran it successfully, and verified the resulting ELF binary as x86-64.



```
us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=us-east-1&connType=standard&iinstanceId=i-041c8c625f1af8dff&osUser=ec2-user&sshPort=22&addressFa...
aws Search [Alt+S]
United States (N. Virginia)
Vivek K K (205542837508)
Vivek%20K%20K

Installed:
annobin-docs-12.69-1.amzn2023.0.1.noarch
annobin-plugin-gcc-12.69-1.amzn2023.0.1.x86_64
cpp-11.5.0-5.amzn2023.0.5.x86_64
gcc-8.0.4-5.amzn2023.0.2.x86_64
gcc-11.5.0-5.amzn2023.0.5.x86_64
gcc-plugin-annobin-11.5.0-5.amzn2023.0.5.x86_64
glibc-devel-2.34-231.amzn2023.0.5.x86_64
glibc-headers-x86-2.34-231.amzn2023.0.5.noarch
guile22-2.2.7-2.amzn2023.0.3.x86_64
kernel6.18-headers-1:6.18.51-120.163.amzn2023.x86_64
libmpc-1.2.1-2.amzn2023.0.2.x86_64
libtool-ltdl-2.4.7-1.amzn2023.0.3.x86_64
libxcrypt-devel-4.4.33-7.amzn2023.x86_64
make-1:4.3-5.amzn2023.0.2.x86_64

Complete!
[ec2-user@ip-172-31-32-199 c-app]$ gcc --version
gcc (GCC) 11.5.0 20240719 (Red Hat 11.5.0-5)
Copyright (C) 2021 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

[ec2-user@ip-172-31-32-199 c-app]$ gcc main.c -o app
[ec2-user@ip-172-31-32-199 c-app]$ ./app
Hello from AWS Multi-Architecture Application!
Architecture: x86_64
[ec2-user@ip-172-31-32-199 c-app]$ file app
app: ELF 64-bit LSB executable, x86-64, version 1 (SYSV), dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2, BuildID[sha1]=61bc7d69653e02b768d4c82b75d51ce276f9
a7ae, for GNU/Linux 3.2.0, not stripped
[ec2-user@ip-172-31-32-199 c-app]$
```

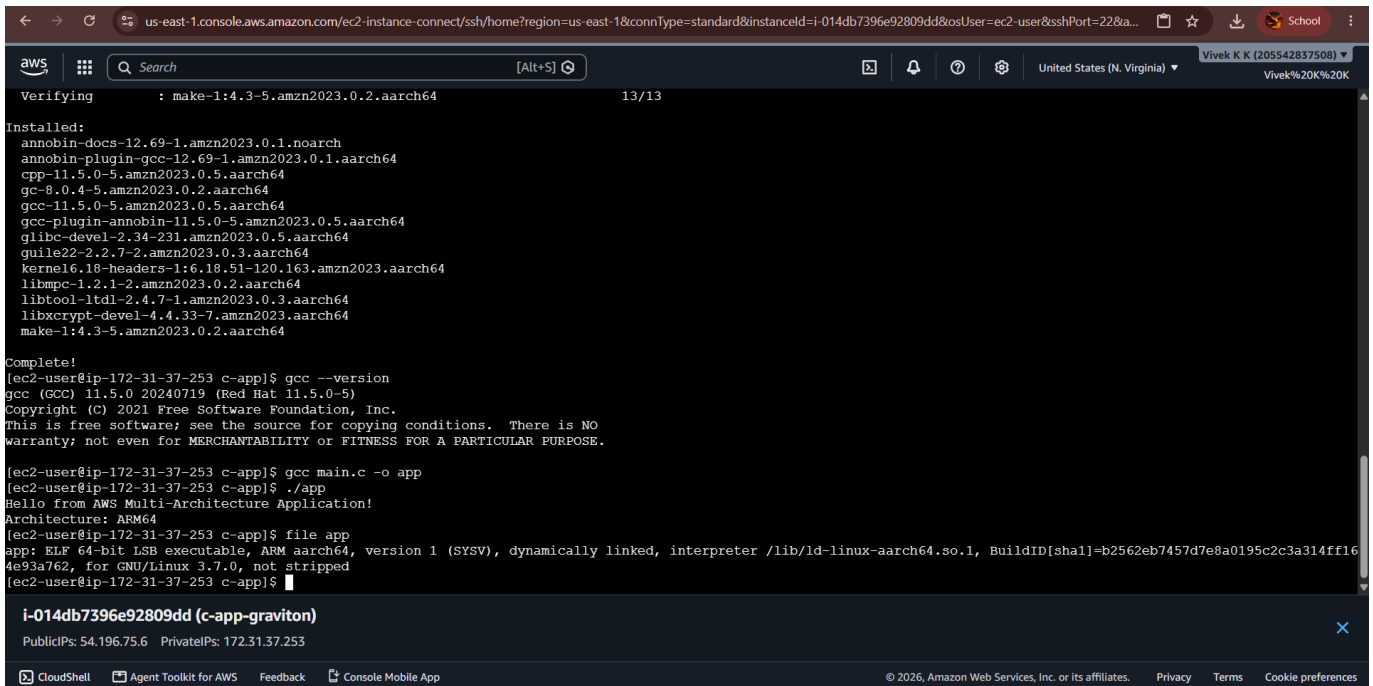
i-041c8c625f1af8dff (c-app-x86)

PublicIPs: 100.30.230.76 PrivateIPs: 172.31.32.199

Figure 9. 9. Install GCC and Build the C Application on x86_64

10. Install GCC and Build the C Application on ARM64

Installed GCC development tools and compiled the same C application on the ARM64 instance, producing an ARM aarch64 executable.



```
Verifying : make-1:4.3-5.amzn2023.0.2.aarch64 13/13

Installed:
annobin-docs-12.69-1.amzn2023.0.1.noarch
annobin-plugin-gcc-12.69-1.amzn2023.0.1.aarch64
cpp-11.5.0-5.amzn2023.0.5.aarch64
gcc-8.0.4-5.amzn2023.0.2.aarch64
gcc-11.5.0-5.amzn2023.0.5.aarch64
gcc-plugin-annobin-11.5.0-5.amzn2023.0.5.aarch64
glibc-devel-2.34-231.amzn2023.0.5.aarch64
guile22-2.2.7-2.amzn2023.0.3.aarch64
kernel6.18-headers-1:6.18.51-120.163.amzn2023.aarch64
libmpc-1.2.1-2.amzn2023.0.2.aarch64
libtool-ltdl-2.4.7-1.amzn2023.0.3.aarch64
libxcrypt-devel-4.4.33-7.amzn2023.aarch64
make-1:4.3-5.amzn2023.0.2.aarch64

Complete!
[ec2-user@ip-172-31-37-253 c-app]$ gcc --version
gcc (GCC) 11.5.0 20240719 (Red Hat 11.5.0-5)
Copyright (C) 2021 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

[ec2-user@ip-172-31-37-253 c-app]$ gcc main.c -o app
[ec2-user@ip-172-31-37-253 c-app]$ ./app
Hello from AWS Multi-Architecture Application!
Architecture: ARM64
[ec2-user@ip-172-31-37-253 c-app]$ file app
app: ELF 64-bit LSB executable, ARM aarch64, version 1 (SYSV), dynamically linked, interpreter /lib/ld-linux-aarch64.so.1, BuildID[sha1]=b2562eb7457d7e8a0195c2c3a314ff16
4e93a762, for GNU/Linux 3.7.0, not stripped
[ec2-user@ip-172-31-37-253 c-app]$
```

i-014db7396e92809dd (c-app-graviton)

PublicIPs: 54.196.75.6 PrivateIPs: 172.31.37.253

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Figure 10. 10. Install GCC and Build the C Application on ARM64

11. Create EC2-to-S3 IAM Role

Created the IAM role EC2-S3-CApp-Role with EC2 as the trusted service, enabling controlled access from the instance to S3.

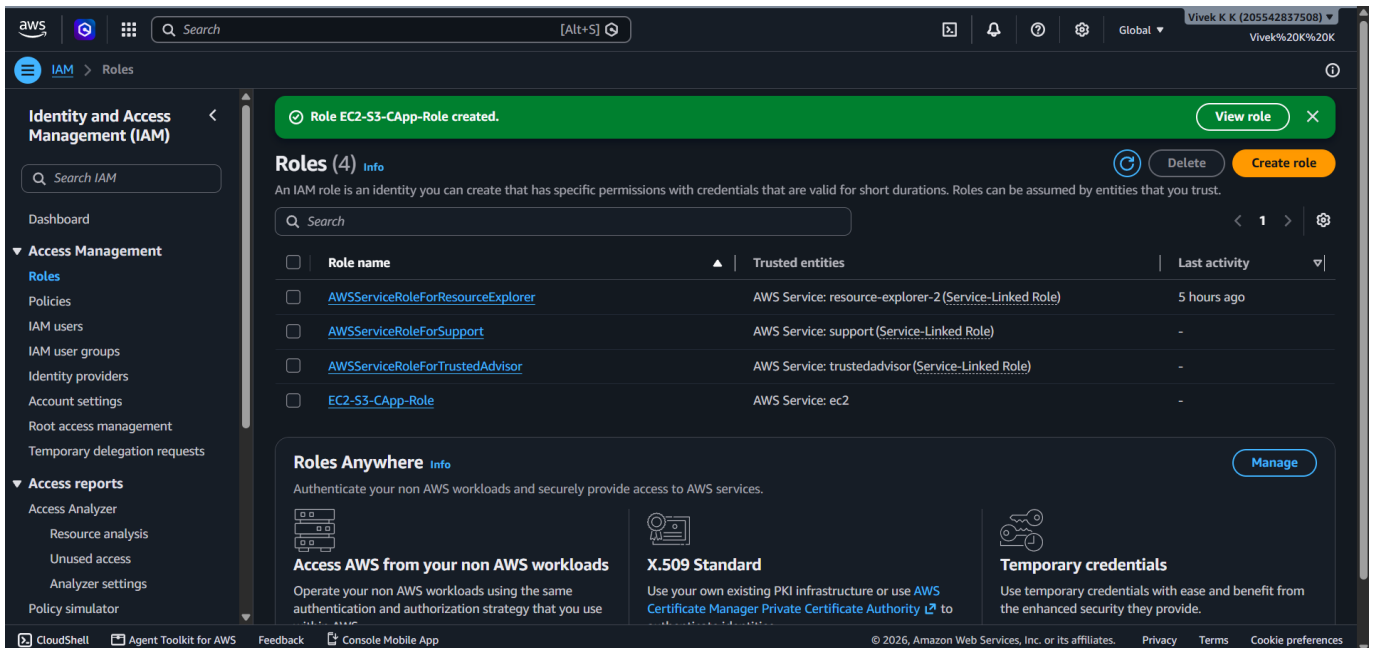
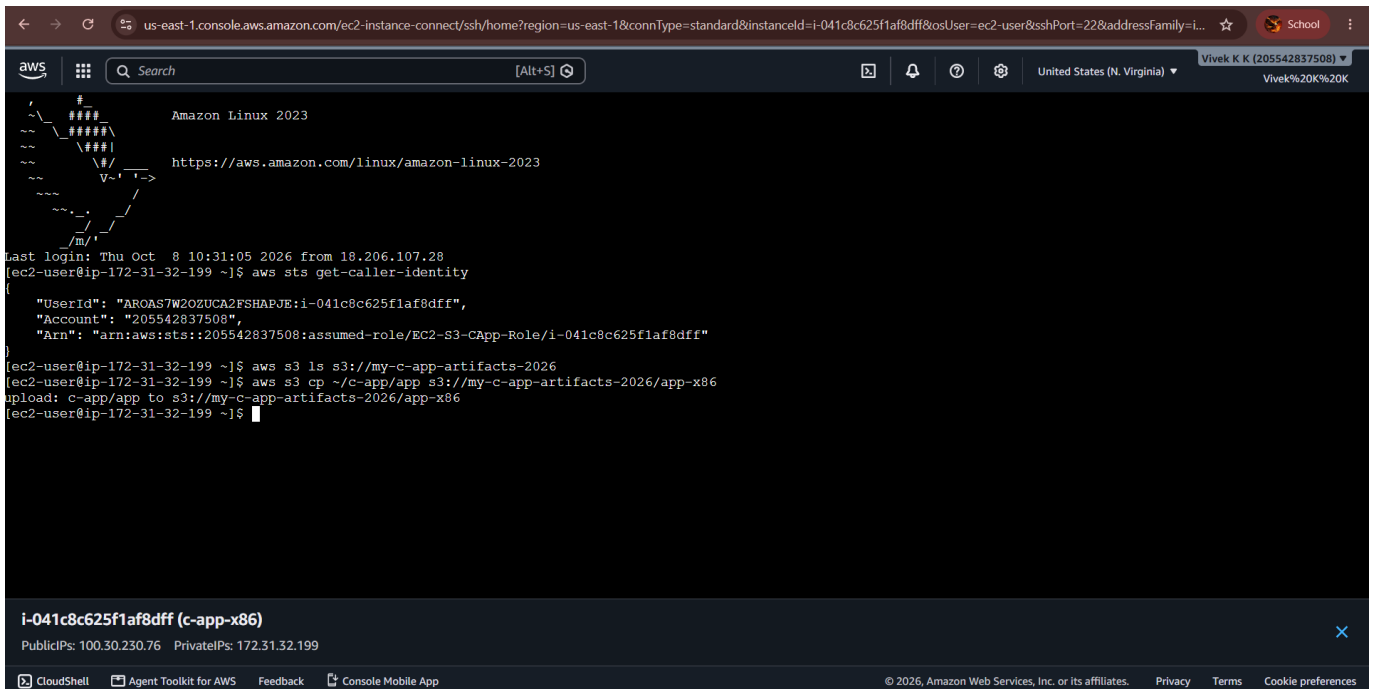


Figure 11. 11. Create EC2-to-S3 IAM Role

12. Verify IAM Role and Upload the x86 Artifact to S3

Verified the assumed IAM role with `aws sts get-caller-identity`, listed the S3 bucket, and uploaded the x86 app binary to S3 as `app-x86`.



The screenshot shows a terminal window within the AWS console. The terminal displays the following commands and output:

```
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

Last login: Thu Oct  8 10:31:05 2026 from 18.206.107.28
[ec2-user@ip-172-31-32-199 ~]$ aws sts get-caller-identity
{
  "UserId": "AROAS7W2OZUCA2FSHAPJE:i-041c8c625f1af8dff",
  "Account": "205542837508",
  "Arn": "arn:aws:sts::205542837508:assumed-role/EC2-S3-CApp-Role/i-041c8c625f1af8dff"
}
[ec2-user@ip-172-31-32-199 ~]$ aws s3 ls s3://my-c-app-artifacts-2026
[ec2-user@ip-172-31-32-199 ~]$ aws s3 cp ~/c-app/app s3://my-c-app-artifacts-2026/app-x86
upload: c-app/app to s3://my-c-app-artifacts-2026/app-x86
[ec2-user@ip-172-31-32-199 ~]$
```

Below the terminal output, a summary box for the instance `i-041c8c625f1af8dff (c-app-x86)` is visible, showing Public IPs: 100.30.230.76 and Private IPs: 172.31.32.199.

Figure 12. 12. Verify IAM Role and Upload the x86 Artifact to S3

13. Create Dockerfile and Start Multi-Architecture Image Build

Created the Dockerfile, checked the local Docker images, and started building the c-multiarch-app image from the GCC base image.

```
aws | [ ] | 🔍 | 📄 | 🔔 | ❓ | ⚙️ | United States (N. Virginia) ▾ | Vivek K K (205542837508) ▾  
Vivek%20K%20K  
"Account": "205542837508",  
"Arn": "arn:aws:sts::205542837508:assumed-role/EC2-S3-CApp-Role/i-041c8c625f1af8dff"  
}  
[ec2-user@ip-172-31-32-199 ~]$ aws s3 ls s3://my-c-app-artifacts-2026  
[ec2-user@ip-172-31-32-199 ~]$ aws s3 cp ~/c-app/app s3://my-c-app-artifacts-2026/app-x86  
upload: c-app/app to s3://my-c-app-artifacts-2026/app-x86  
[ec2-user@ip-172-31-32-199 ~]$ cd ~/c-app  
[ec2-user@ip-172-31-32-199 c-app]$ nano Dockerfile  
[ec2-user@ip-172-31-32-199 c-app]$ sudo docker images  

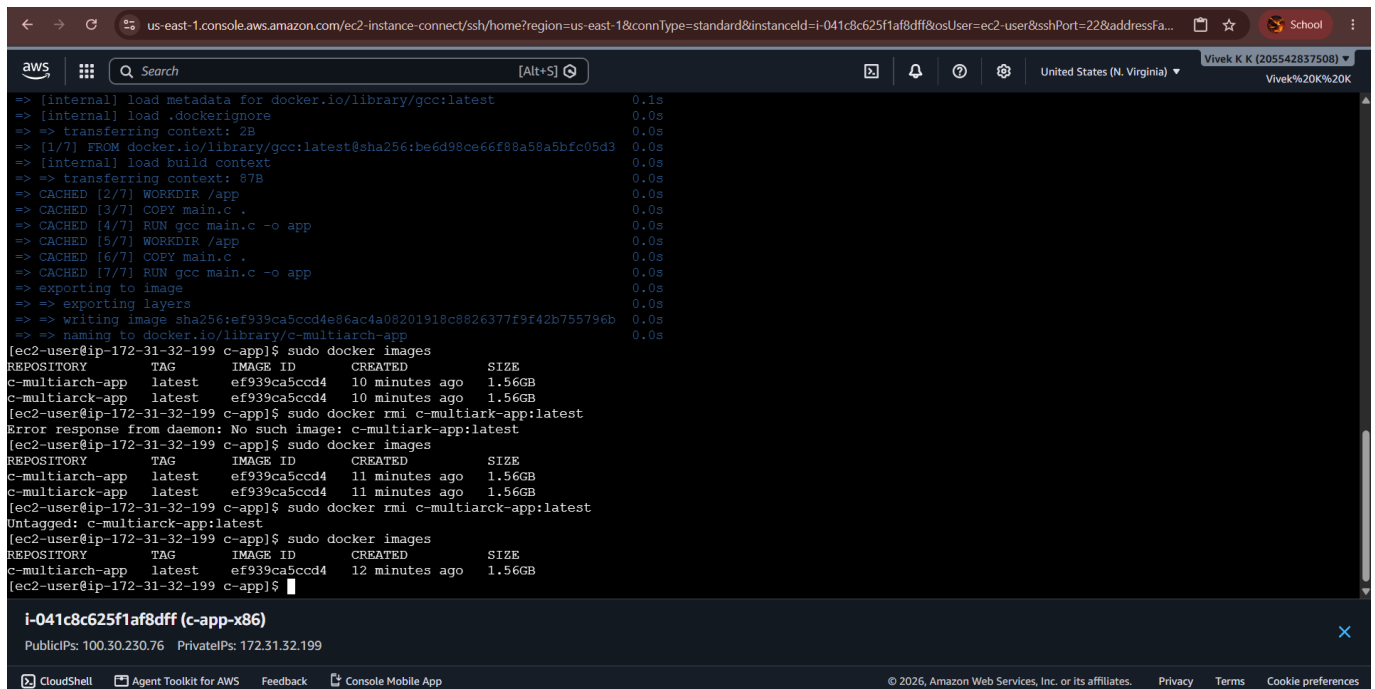

| REPOSITORY | TAG | IMAGE ID | CREATED | SIZE |
|------------|-----|----------|---------|------|
|------------|-----|----------|---------|------|

  
[ec2-user@ip-172-31-32-199 c-app]$ sudo docker build -t c-multiarch-app .  
[+] Building 1.1s (4/11)                                docker:default  
=> [internal] load build definition from Dockerfile      0.1s  
=> => transferring dockerfile: 263B                      0.0s  
=> [internal] load metadata for docker.io/library/gcc:latest 0.5s  
=> [internal] load .dockerignore                        0.0s  
=> => transferring context: 2B                            0.0s  
=> [1/7] FROM docker.io/library/gcc:latest@sha256:be6d98ce66f88a58a5bfc05d3 0.3s  
=> => resolve docker.io/library/gcc:latest@sha256:be6d98ce66f88a58a5bfc05d3 0.0s  
=> => sha256:be6d98ce66f88a58a5bfc05d3a81925ae01e3164dc89a7 7.63kB / 7.63kB 0.0s  
=> => sha256:3f73339bf0d3d64081cac3f6d9fdafc6dd4cef2b9c49e0 2.49kB / 2.49kB 0.0s  
=> => sha256:62fe2e747f69f2f414c7e9eb1553155cf0a658398f04 16.78MB / 25.64MB 0.2s  
=> => sha256:959ac766250f0fc0178b0af7f4454756bd3c16c6241af7 7.46kB / 7.46kB 0.0s  
=> => sha256:4846f29007305297af653c57bf9b60ba8f676a38695c6 6.29MB / 49.39MB 0.2s  
=> => sha256:4dc3950125ebc9ed195e1101fc6c02e2fce113ba0f53 15.73MB / 67.81MB 0.2s  
=> [internal] load build context                          0.0s  
=> => transferring context: 381B                             0.0s
```

Figure 13. 13. Create Dockerfile and Start Multi-Architecture Image Build

14. Complete Docker Image Build and Verify Image

Completed the Docker build successfully and verified the c-multiarch-app:latest image in the local Docker image list.



```
us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=us-east-1&connType=standard&instanceId=i-041c8c625f1af8dff&osUser=ec2-user&sshPort=22&addressFa...
AWS [Search] [Alt+S] United States (N. Virginia) Vivek K K (205542837508) Vivek%20K%20K

-> [internal] load metadata for docker.io/library/gcc:latest 0.1s
-> [internal] load .dockerignore 0.0s
-> => transferring context: 2B 0.0s
-> [1/7] FROM docker.io/library/gcc:latest@sha256:be6d98ce66f80a58a5bfc05d3 0.0s
-> [internal] load build context 0.0s
-> => transferring context: 87B 0.0s
-> CACHED [2/7] WORKDIR /app 0.0s
-> CACHED [3/7] COPY main.c . 0.0s
-> CACHED [4/7] RUN gcc main.c -o app 0.0s
-> CACHED [5/7] WORKDIR /app 0.0s
-> CACHED [6/7] COPY main.c . 0.0s
-> CACHED [7/7] RUN gcc main.c -o app 0.0s
-> exporting to image 0.0s
-> => exporting layers 0.0s
-> => writing image sha256:ef939ca5ccd4e86ac4a08201918c8826377f9f42b755796b 0.0s
-> => naming to docker.io/library/c-multiarch-app 0.0s
[ec2-user@ip-172-31-32-199 c-app]$ sudo docker images
REPOSITORY          TAG             IMAGE ID        CREATED         SIZE
c-multiarch-app     latest          ef939ca5ccd4   10 minutes ago 1.56GB
c-multiarch-app     latest          ef939ca5ccd4   10 minutes ago 1.56GB
[ec2-user@ip-172-31-32-199 c-app]$ sudo docker rmi c-multiarch-app:latest
Error response from daemon: No such image: c-multiarch-app:latest
[ec2-user@ip-172-31-32-199 c-app]$ sudo docker images
REPOSITORY          TAG             IMAGE ID        CREATED         SIZE
c-multiarch-app     latest          ef939ca5ccd4   11 minutes ago 1.56GB
c-multiarch-app     latest          ef939ca5ccd4   11 minutes ago 1.56GB
[ec2-user@ip-172-31-32-199 c-app]$ sudo docker rmi c-multiarch-app:latest
Untagged: c-multiarch-app:latest
[ec2-user@ip-172-31-32-199 c-app]$ sudo docker images
REPOSITORY          TAG             IMAGE ID        CREATED         SIZE
c-multiarch-app     latest          ef939ca5ccd4   12 minutes ago 1.56GB
[ec2-user@ip-172-31-32-199 c-app]$
```

i-041c8c625f1af8dff (c-app-x86)
PublicIPs: 100.30.230.76 PrivateIPs: 172.31.32.199

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Figure 14. 14. Complete Docker Image Build and Verify Image